

Student Performance Analytics and AI-Based Prediction

Internship Project Report

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Project domain	Data Analytics / Machine Learning
Tools	Python, Pandas, Matplotlib, Scikit-learn
Dataset	Synthetic demonstration dataset (40 records)
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A practical demonstration of how data analysis and machine learning can support early academic intervention. All included records are synthetic and used only for demonstration.

1. Abstract

This project demonstrates a compact data analytics and artificial intelligence workflow for exploring factors associated with student academic performance. It uses a synthetic dataset of 40 student-like records containing study hours, attendance, assignment score, quiz score, sleep hours and final score. Python is used for data inspection, descriptive statistics, visualization, support-category summaries and a Random Forest regression model. The project illustrates a workflow rather than a validated educational product; its small synthetic dataset cannot establish real-world causal relationships or reliable predictions.

2. Problem Statement

Educators often need a clear, evidence-informed way to review learning indicators and identify where additional academic support may be useful. The project asks: (1) What patterns appear in the sample data? (2) Can a basic machine-learning model estimate final scores from selected learning indicators? (3) How can results be communicated responsibly?

3. Objectives

- Inspect dataset structure and missing values.
- Summarize key variables with descriptive statistics.
- Visualize study time and score patterns.
- Build and evaluate a Random Forest regression baseline.
- Generate an illustrative support category and explain limitations.

4. Dataset and Tools

The bundled CSV contains 40 synthetic records. Features are study hours per day, attendance percentage, assignment score out of 10, quiz score out of 10 and sleep hours. The target variable is final score percentage. Tools: Python 3, Pandas for data handling, Matplotlib for plots, and scikit-learn for train/test splitting, Random Forest regression and evaluation.

Data dictionary

Column	Meaning
Student_ID	Anonymous demo identifier
Study_Hours_Per_Day	Daily study time in hours
Attendance_Percent	Attendance percentage
Assignment_Score_10	Assignment score out of 10
Quiz_Score_10	Quiz score out of 10
Sleep_Hours	Reported sleep hours in demo data
Final_Score_Percent	Final score percentage (target)

5. Methodology

Step 1 — Data loading

Load the CSV with Pandas and inspect rows, dimensions, data types, missing values and summary statistics.

Step 2 — Exploratory data analysis

Create a scatter plot of study hours versus final score and a bar chart of students in three illustrative support categories.

Step 3 — Prepare model inputs

Use five selected numeric features to predict final score. Split data into training and held-out test sets using a fixed random seed for reproducibility.

Step 4 — Train the model

Fit a Random Forest Regressor with 200 trees. This non-linear ensemble model can capture interactions in tabular data.

Step 5 — Evaluate and report

Report Mean Absolute Error (MAE) and R-squared on the held-out demo subset, and export actual-versus-predicted test results.

Step 6 — Responsible interpretation

Treat the model output as an estimate only. Never use it alone to make high-impact decisions about students; investigate data quality, bias, privacy and generalization first.

6. Expected Analysis and Interpretation

Because the dataset is deliberately constructed for demonstration, it is designed to show a plausible positive association between study-related indicators and final scores. This is not evidence that study hours alone cause higher marks. Attendance, prior knowledge, teaching quality, assessment design, personal circumstances and many other factors may influence outcomes. Run the supplied Python script to generate the actual descriptive statistics, plots, feature importance and evaluation metrics for this exact CSV.

7. AI Model

The Random Forest Regressor combines predictions from many decision trees. It is suitable as an accessible baseline for tabular regression. The data are split into 75% training and 25% test records. MAE expresses the average absolute prediction error in score percentage points; R-squared summarizes fit relative to a simple mean baseline, but can be unstable on a small test set. Metrics from this tiny synthetic dataset must not be presented as proof of production accuracy.

8. Deliverables

The project package includes: (a) this report, (b) the runnable Python source file, (c) the synthetic CSV dataset, and (d) a README with run instructions. Running the script saves plots and a CSV of test predictions in a `project_outputs` folder.

9. Limitations and Ethical Considerations

- Dataset is synthetic and very small; it is not representative of a real school or college.
- A random split of 40 rows produces uncertain performance estimates.

- The sample patterns are not causal evidence.
- Real student data require permission, data minimization, secure storage and appropriate consent.
- Support flags should prompt supportive conversation, not stigmatize students or determine opportunities.
- Before deployment, validate on larger, diverse, consented data and compare with simpler baselines.

10. Future Scope

Future improvements include collecting approved anonymized data, adding subject-wise performance and previous-term trends, comparing linear regression with Random Forest, using cross-validation, monitoring errors across student groups, and developing a simple dashboard with filters and clear explanations. Any real-world version should be co-designed with educators and students.

11. Conclusion

This project demonstrates the end-to-end structure of a beginner-friendly Data Analytics and AI project: data preparation, exploratory analysis, visualization, supervised learning, evaluation and responsible communication. Its main value is as a reproducible learning prototype. The supplied code allows the reviewer to rerun the analysis and inspect outputs rather than relying only on screenshots or unsupported claims.

12. References

- McKinney, W. — Python for Data Analysis, O'Reilly Media.
- Pandas documentation: <https://pandas.pydata.org/docs/>
- Matplotlib documentation: <https://matplotlib.org/stable/>
- Scikit-learn documentation: <https://scikit-learn.org/stable/>

Declaration: This report and dataset are a demonstration project. Replace the student details if required by the internship, and follow the provider's originality and submission rules.